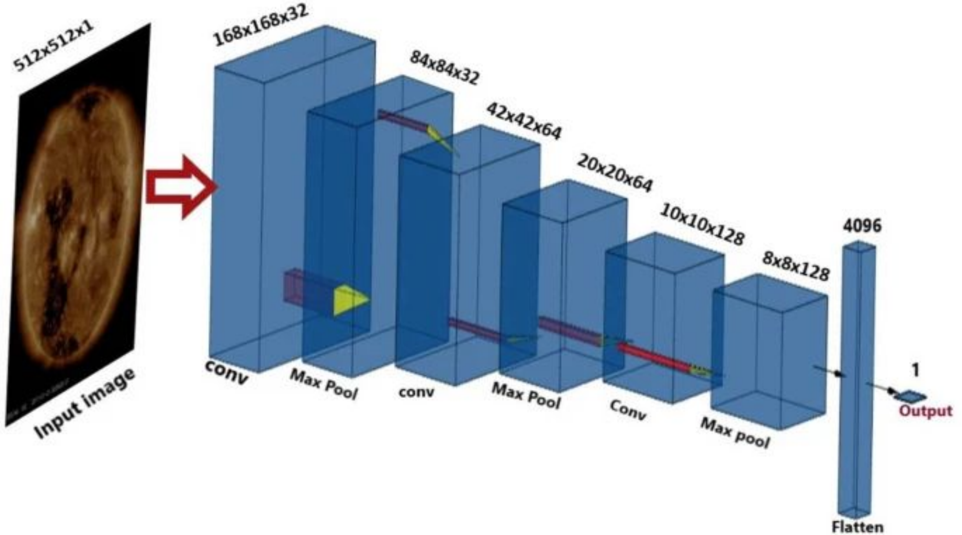
Solar wind speed prediction

Raju, H., Das, S. CNN-Based Deep Learning Model for Solar Wind Forecasting. Sol Phys 296, 134 (2021)

Single SDO 193A image, solar wind speed at 1 AU by ACE. Lead time is 1AU divided by predicted v. 2013 – 15 and 2017 for training and validation, and 2018 for testing.



Reported performance

Model	Analysed period	CC	RMSE [$\mathrm{km\,s^{-1}}$]
Proposed CNN	2018	0.57 ± 0.02	76.3 ± 1.87
1d-persistence	2018	0.62	77
PS-27 (Owens et al., 2013)	2018	0.45	94
CH model (Rotter et al., 2015)	2018	0.38	102
Constant four-day delay	2018	0.55 ± 0.02	78 ± 3.6

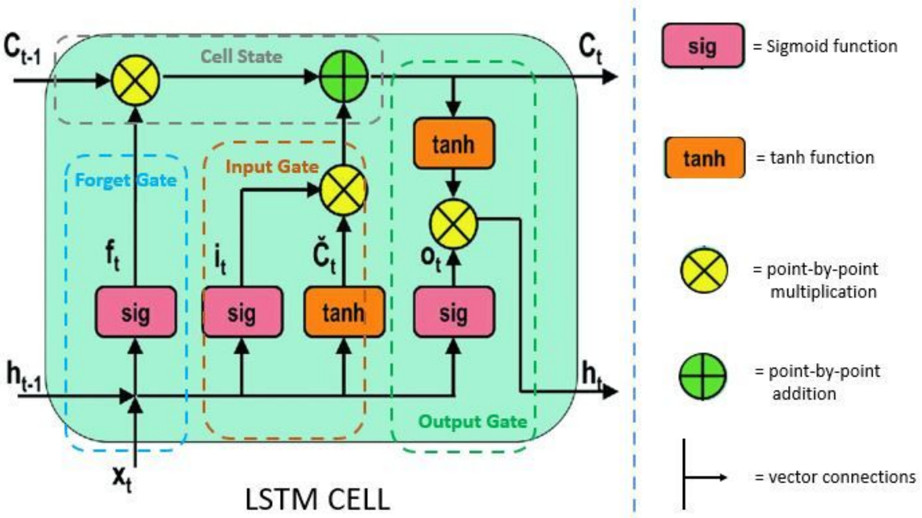
Dynamic lead time produced slight better performance than a fixed four-day lead, but not that the choice of exactly 4-days may not be optimal, thought it would be expensive to investigate.

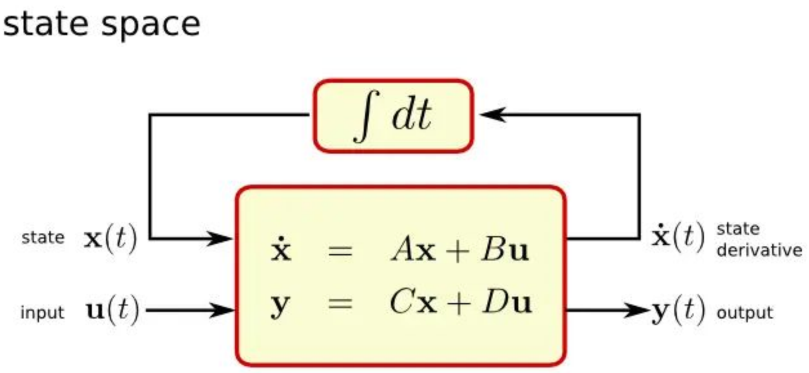
Predicting using the constant average gives RMSE ~82.

Rmse high. Only slightly better than a constant predictor. This suggests that the solar wind around earth is not determined by a single instance of the solar energy signature. The sun releases plasmas continuously with varying initial velocity, and those plasma interact in the their trajectory towards earth, so the simple ballistic model cannot model this well.

On the other hand, it is reasonable to assume that solar wind speed is primarily determined by solar energy signal over a relatively short window: within days if not hours. Two possibly approaches next:

1: Follow feature extractors by LSTM/SSM, by which we hope to learn the accumulating effect of solar emission on solar wind near earth.





System represented as a collection of coupled linear first-order differential equations.

2: Physics informed approach. How plasma/CME interact during the trajectory, intuitively they can be approximated by differential equations, while SDO acts like boundary conditions/input.

Then a natural thought would to investigate whether initial state matters. A good starting point would be the initial solar wind speed.

Carrington rotation: May used in data split?

PINN excels in approximating differential equations with unknown coefficients, need to look for literatures modeling the solar wind propagation.

Surya: Vision transformer, input is a single time instance of all channels of AIA and HMI, with 4k resolution. It would be reasonable to expect it to be an effective feature extractor tuned for space weather.

Sunspot groups: state of solar cycle

Test case to be with cv or a separate year.

Computing power

Reported results from Raju and my test results

Raju: 80%/20% training and validation split on combined data in 2013-2015 and 2017, and testing on 2018, with both dynamic delta t (1AU/predicted speed) and a fixed 4 days time lead

Me: 80%/20% training and validation split on 2018 and tested on 2018

Method	RMSE
Raju test with 4 days lead	78
Raju validation with dynamic lead	82
Raju test with dynamic lead	76.3
My validation with 4 days lead	0.03 normalized and squared
My test with 4 days lead	67.37
My validation with dynamic lead	0.0211
My test with dynamic lead	63.92
1 day persistence	77
27 days persistence	94

Method	RMSE
Raju test with 4 days lead	78
Raju test with dynamic lead	76.3
My test with 4 days lead	92.43
My test with dynamic lead	102.9
1 day persistence	77
27 days persistence	94